Regional medullary redox indices and *Nox1*/*Nox4* mRNA expression in rats with atlantoaxial joint disorder

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Summary

We examined regional medullary redox indices and *Nox1*/*Nox4* transcripts in male Wistar rats with atlantoaxial joint disorder. Sham-operated and model rats were sampled at week 4; restoration rats underwent implant removal at week 4 and sampling at week 8. Each endpoint included six independent rats per group. In tissue containing the rostral ventrolateral medulla (RVLM) and adjacent medulla, week-4 model rats had lower SOD activity, higher MDA and probe-associated fluorescence, and estimated *Nox1*/*Nox4* expression of 1.46/2.90-fold relative to sham rats. Restoration values were generally closer to sham values at the later endpoint. Unmatched sampling times, absent second-surgery controls, unvalidated tissue fluorescence-assay applicability, incomplete qPCR quality-control records and omission of postoperative analgesia limit interpretation. These exploratory observations do not establish RVLM-specific changes, NOX-dependent sympathetic mediation or an implant-removal effect.

Key words: Atlantoaxial joint disorder; Ventrolateral medulla; Oxidative stress; NADPH oxidase; RT-qPCR

A metallic implant in the rat atlantoaxial joint produces misalignment and blood-pressure elevation [1]. Oxidative stress is implicated in hypertension [2]; the rostral ventrolateral medulla (RVLM) regulates blood pressure [3], and experimental studies link regional redox disturbances to cardiovascular regulation [4,5]. We examined medullary biochemical indices and *Nox1*/*Nox4* transcripts in this model. Blood-pressure observations were shared with a related Ang II/AT1-R report [6], whereas the present biochemical and transcript analyses were not reported there.

The Animal Ethics Committee of Anhui University of Chinese Medicine approved the 60-rat parent experiment, model induction, implant removal and terminal sampling (AHUCM-rats-2022032). Clean-grade male Wistar rats (approximately 12 weeks, 250 g; Anhui Medical University, license SCXK [Wan] 2017-001) were individually housed at 22 ± 1°C and 60% humidity under 12 h light/dark cycles, with free food/water and 1-week acclimatization.

Animals were randomized to sham operation (10) or model induction (50). Six rats died intraoperatively and three afterward; seven survivors lost implants, leaving 34 with retained implants. Successfully modeled rats were randomized to model, restoration or superior cervical ganglion resection groups (10 each); four remained in reserve. Restoration rats continued from this successfully modeled cohort to week 8, while separate model rats underwent terminal sampling at week 4. This report includes sham, model and restoration cohorts only, initially 10 per group; each biochemical endpoint and RT-qPCR used six randomly selected independent rats per group without pooling. No single primary endpoint or formal a priori power calculation was specified. Animal-to-sample identifiers and cross-assay overlap could not be recovered.

Surgery used 2% inhaled isoflurane at 1 l/min, with respiratory, righting-reflex and noxious-withdrawal monitoring. An S-shaped aluminum implant was placed in the left atlantoaxial joint through an anterior cervical approach [1]; sham rats underwent exposure without implantation. Week-4 screening required retained implantation, joint asymmetry with a left–right odontoid–lateral-mass distance difference >0.05 mm, and systolic pressure ≥20% above baseline with a week-4 plateau. Restoration rats underwent implant removal at week 4. Penicillin sodium was given intramuscularly at 400,000–800,000 units/kg daily for 3 days. Possible contributors to the nine deaths were anesthetic overdose, cervical injury and infection; individual cause-of-death adjudication records were unavailable. No postoperative analgesics were administered; no written rationale or specific ethics-committee opinion about this omission was retained. Food intake, activity and wounds were checked daily for 3 days and 2–3 times weekly thereafter.

Sham/model rats were sampled at week 4 and restoration rats at week 8, without a time-matched second-surgery sham control. Abdominal aortic sampling under isoflurane was followed by inhaled-isoflurane overdose; death was confirmed when spontaneous breathing, palpable cardiac activity and withdrawal to a noxious stimulus were all absent, before brainstem removal. Atlas-guided bilateral tissue blocks (approximately 100 mg/rat) contained RVLM and adjacent medulla. For biochemical assays, tissue was lysed in RIPA/PMSF and protein measured by BCA. SOD (A001-3, WST-1) and MDA (A003-1, thiobarbituric acid; Nanjing Jiancheng) used RT-6000 absorbance readings at 450 and 530 nm, respectively; the MDA manual specified 532 nm. The fluorescence-reader model was not retained. BestBio BB-475015 O12 fluorescence was measured at excitation/emission 488/530 nm and divided by protein concentration, yielding an arbitrary-unit (AU) index. This serum-designated assay lacked tissue-matrix and buffer-compatibility validation; fluorescence is not an absolute ROS concentration or species-specific measure.

RNA extraction used TRIzol; reverse transcription used PrimeScript RR047A, followed by SYBR RT-qPCR on PikoReal 96. Each target and paired *β-actin* reference was measured once per rat, without technical replication. Within-rat ΔCq was target minus reference Cq. Relative quantity was calculated as $Q = 2^{-\Delta Cq}$ and divided by the arithmetic mean quantity of the six sham rats [7]. This assumes equal, approximately 100% amplification efficiencies. Original Cq, amplification and melting exports are provided in Supplementary File 7. Target melt profiles were heterogeneous, and two wells had conflicting plate/result annotations. Efficiencies and β-actin stability were not validated; no-template/no-reverse-transcription control records were unavailable. The six supplied records per group were retained unchanged for exploratory reporting. Reagent details, primers, protocols and source-flow information are retained in Supplementary Methods.

Individual rats were experimental units; results are mean ± sample SD. All six supplied values per group were included, without outlier removal or imputation. Biochemical

comparisons used Welch ANOVA and Games–Howell tests, adjusting three comparisons within each endpoint, without adjustment across endpoints. RT-qPCR inference used Welch t tests on ΔCq, with Holm correction across all 18 comparisons in the original six-target panel, not only the displayed genes. Tests were two-sided ($\alpha = 0.05$). Inputs, complete outputs, confidence intervals and code accompany the manuscript; cross-assay correlations were not tested.

Across the terminal cohorts, Welch ANOVA gave $F(2, 9.816) = 91.626$ for SOD, $F(2, 8.500) = 130.738$ for MDA and $F(2, 7.804) = 314.907$ for fluorescence (all $P < 0.001$). In week-4 model versus sham rats, SOD was 0.564 ± 0.072 versus 1.122 ± 0.072 U/mg protein, MDA 0.349 ± 0.034 versus 0.095 ± 0.017 nmol/mg protein, and fluorescence 3295.6 ± 216.3 versus 1326.9 ± 50.6 AU (Fig. 1; all Games–Howell-adjusted $P < 0.001$; Supplementary Table S1). Estimated *Nox1* expression was 1.459 ± 0.156 versus 1.000 ± 0.134, and *Nox4* 2.902 ± 0.558 versus 1.000 ± 0.233 (Fig. 2; Holm-adjusted $P = 0.0037$ and 0.0001).

At the later restoration endpoint, SOD was 1.035 ± 0.101, MDA 0.213 ± 0.049 and fluorescence 2265.6 ± 131.6 in the same respective units; all differed from model values (adjusted $P < 0.001$). Relative to sham, MDA and fluorescence remained higher ($P = 0.0033$ and <0.001), whereas the SOD difference was nonsignificant ($P = 0.2534$). Restoration *Nox1*/*Nox4* estimates were 1.180 ± 0.090/1.645 ± 0.227, lower than model values (adjusted $P = 0.0139/0.0037$). Compared with sham, *Nox4* remained higher ($P = 0.0099$); *Nox1* was nonsignificant ($P = 0.0618$). Nonsignificance does not establish equivalence or normalization. Full three-cohort statistics are supplied.

Lower SOD activity and higher MDA provide complementary regional antioxidant and lipid-peroxidation observations [8]. Together with estimated *Nox1*/*Nox4* elevations, these findings extend the model's characterization [1] and are consistent with redox disturbances in other hypertension models [4,5]. Transcript abundance does not establish increased NOX protein, oxidase activity or sympathetic output. Unvalidated O12 fluorescence is auxiliary; common processing cannot exclude differential matrix effects. Archived H&E fields and blood-pressure records are descriptive only (Supplementary Figs. S1/S2), with unconfirmed histological animal counts and unreconstructed terminal blood-pressure cohorts.

Interpretation remains constrained by regional rather than RVLM-specific sampling, heterogeneous target melt profiles, conflicting qPCR well annotations, incomplete validation and untraceable cross-assay identities. Later restoration observations confound implant removal with age, elapsed time, recovery and repeat anesthesia/surgery. Omission of postoperative analgesia raises animal-welfare concerns and possible pain/stress confounding. Small exploratory samples and male-only inclusion limit precision and generalizability. The same-day model–sham comparison supports a regional phenotype; validated assays, anatomical localization and time-matched controls are required to investigate NOX-dependent sympathetic mediation or an implant-removal effect.

Acknowledgments

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Conflict of Interest

The authors declare no competing interests.

Author contributions

Xiangying Yuan: Conceptualization, Investigation, Formal analysis, Writing – original draft. Bin Yang: Methodology, Investigation. Zongbao He: Investigation, Resources. Youkui Lv: Conceptualization, Supervision, Funding acquisition, Writing – review & editing. Zijian Wu: Conceptualization, Supervision, Project administration, Funding acquisition, Writing – review & editing.

Data availability

Inputs, complete statistical outputs/code, detailed methods, Table S1, descriptive archived H&E/blood-pressure records and original RT-qPCR exports/curve images accompany the manuscript. The RT-qPCR exports reproduce the supplied Cq values but do not resolve the well-annotation or specificity concerns. Complete validation records and animal-to-sample identifiers remain unavailable.

DECLARATION on the Use of AI

Codex (OpenAI; final-revision CLI 0.162.0-alpha.17.2, ChatGPT desktop 26.1007.30912; October 2026; underlying model identifier unarchived) assisted with editing, organization, methodological review, worksheet checking, statistical reanalysis and conventional analysis/plotting code. Authors verified the outputs and take responsibility. No experimental data or images were generated.

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Figure 1

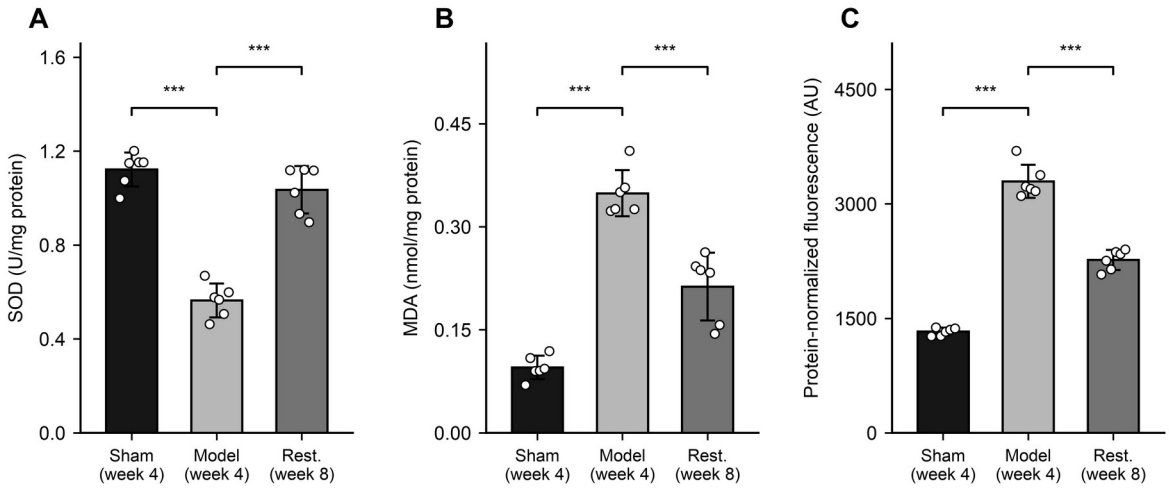


Fig. 1. Regional medullary SOD activity (A), MDA (B) and probe-associated fluorescence (C). Bars show mean ± sample SD; points indicate six independent rats/group. Brackets compare model with sham or restoration; ***P < 0.001 (Games–Howell after Welch ANOVA; three comparisons/endpoint). Fluorescence is protein-concentration-normalized AU. Sham/model: week 4; restoration: week 8. Sham, sham-operated; Rest., restoration. Full outputs: Supplementary File 3.

Figure 2

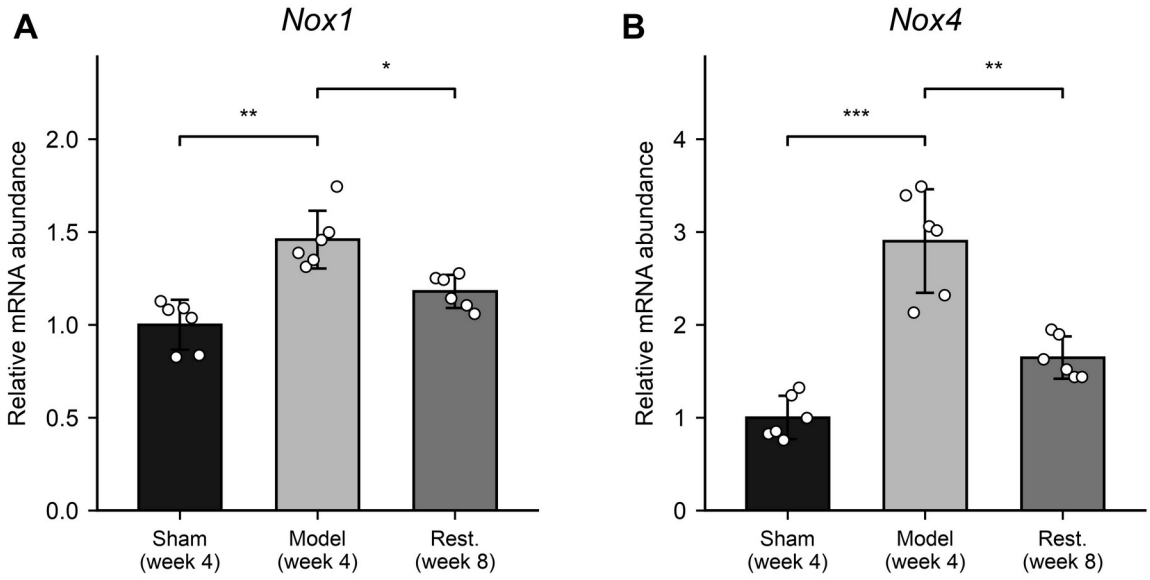


Fig. 2. Relative *Nox1* (A)/*Nox4* (B) transcripts; mean ± sample SD and six independent rats/group. Each rat's quantity was normalized to *β-actin* and the sham arithmetic mean. Brackets compare model with sham or restoration. Stars denote Holm-adjusted P values: *P < 0.05, **P < 0.01, ***P < 0.001, from Welch tests on ΔCq across all 18 original-panel comparisons. Sham/model: week 4; restoration: week 8. Sham, sham-operated; Rest., restoration. Full outputs: Supplementary File 4.